

23. Radiation heat transfer between blackbodies

HT between BBs

$q_{i \rightarrow j}$: Radiation HT from surface i to j

$q_{j \rightarrow i}$: Radiation HT from surface j to i

$q_{ij} \equiv q_{i \rightarrow j} - q_{j \rightarrow i}$: Net radiation heat transfer

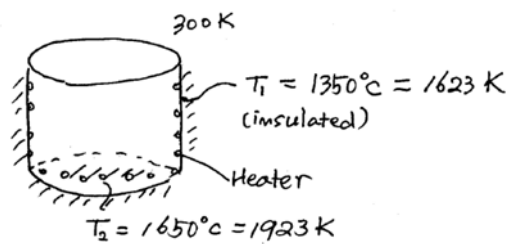
Intuitively,

$$q_{ij} = (A_i J_i) F_{ij} = A_i F_{ij} E_{bi}$$

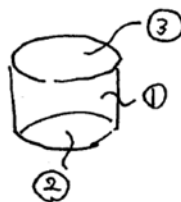
$$q_{ji} = (A_j J_j) F_{ji} = A_j F_{ji} E_{bj} = A_i F_{ij} E_{bj} \quad \because A_i F_{ij} = A_j F_{ji} \text{ (reciprocity)}$$

$$q_{ij} = q_{i \rightarrow j} - q_{j \rightarrow i} = A_i F_{ij} (E_{bi} - E_{bj}) = A_i F_{ij} \sigma (T_i^4 - T_j^4)$$

Ex)



How much heater power to maintain the surface temp?
(Heat loss?)



$$q = q_{13} + q_{23} \quad ?$$

$$q_{13} = A_1 F_{13} \sigma (T_1^4 - T_3^4)$$

$$q_{23} = A_2 F_{23} \sigma (T_2^4 - T_3^4)$$

From table 13.2 or Fig 13.5, $F_{23} = 0.06$

$$F_{31} = 1 - F_{23} = 0.94$$

$$F_{12} = \frac{A_2}{A_1} F_{21} = 0.118 \text{ (by reciprocity)}$$

$$F_{13} = F_{12} = 0.118 \text{ (by symmetry)}$$

$$\therefore q = 1639 + 205 = 1844 \text{ W}$$